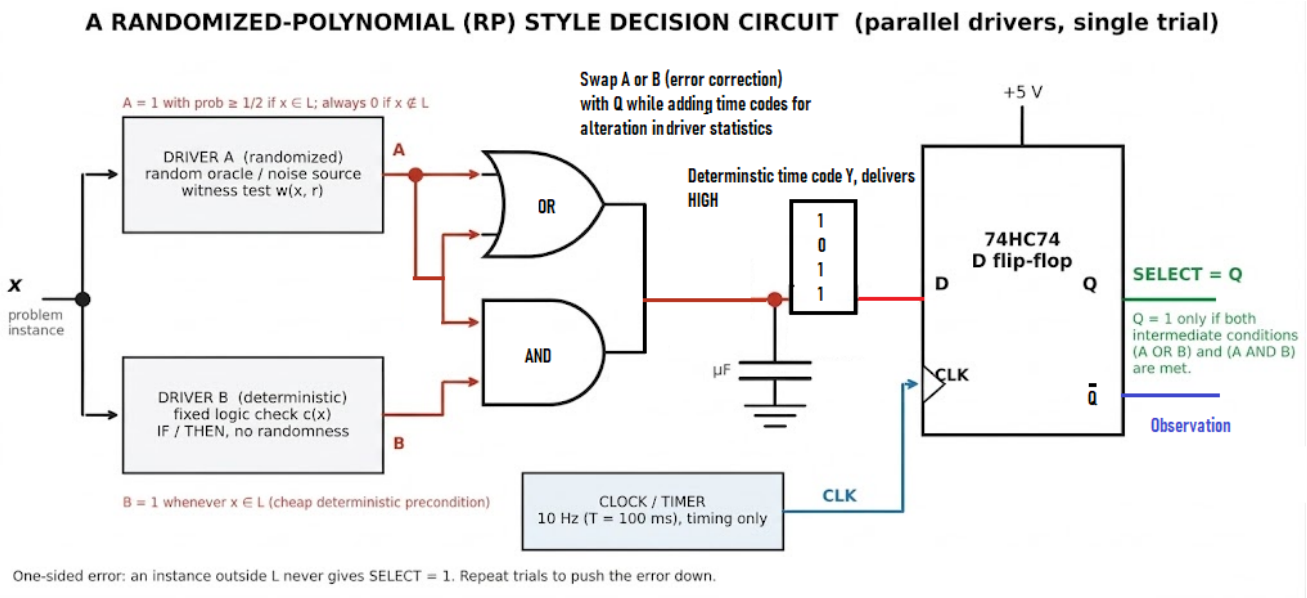
How the RP-Style Decision Circuit Works

A walkthrough of the parallel-driver, single-trial circuit (A, B, OR/AND, time code Y, 74HC74 flip-flop)



The circuit as supplied

1. The big idea

The circuit is a hardware picture of a **Randomized Polynomial-time (RP)** decision procedure. RP is the class of yes/no problems for which a probabilistic algorithm has **one-sided error**: if the answer is **NO** (x is not in L) the algorithm *never* says yes, and if the answer is **YES** (x is in L) it says yes with probability at least 1/2. The circuit runs one such trial. It takes the problem instance **x**, runs it through two drivers in parallel, combines their outputs with logic gates, and stores the verdict in a flip-flop on a clock edge. That stored bit, **SELECT = Q**, is the answer.

2. The blocks and what each one does

Block	Role	Behaviour
Driver A (randomized)	Probabilistic witness test	Uses a random oracle / noise source to try a random witness r and test w(x, r). If x is in L, A = 1 with probability ≥ 1/2. If x is not in L, A = 0 always (no witness can exist).
Driver B (deterministic)	Cheap precondition	A fixed IF/THEN check c(x) with no randomness. B = 1 whenever x is in L. It acts as a sanity filter that costs almost nothing.
OR gate	“Swap / error-correction” branch	Outputs 1 when A or B (or both) is 1, so either driver can contribute. In the figure's notes it is the stage that swaps A or B and adds time codes to alter the driver statistics.
AND gate	Agreement check	Outputs 1 only when A = 1 and B = 1, i.e. the random witness test succeeded *and* the deterministic check agrees.
Capacitor (μF) node	Smoothing / hold	The gate outputs meet at a node with a capacitor to ground. It low-pass filters glitches so the signal is stable when the clock samples it.
Time code Y = 1011	Deterministic time code	A fixed bit pattern that delivers a HIGH to the D input, giving the stage a known, repeatable timing signature.
Clock / timer	Sampling instant	10 Hz (T = 100 ms), “timing only”. It carries no data, it just tells the flip-flop when to look at D.

Block	Role	Behaviour
74HC74 D flip-flop	Memory of the verdict	On the rising CLK edge it copies D to Q and holds it. Q is SELECT (the decision). Q̄ (Q-bar) is the inverted output, labelled Observation.

3. Step-by-step signal flow

1. **Input.** The instance x is fanned out to both drivers at once (the black dot at the left).
2. **Drivers fire in parallel.** A flips its random coin(s) and tests a witness. B runs its fixed check. Both produce one bit, labelled A and B in red.
3. **Gates combine them.** A and B go to the OR and AND gates. The caption summarises the requirement as two intermediate conditions: (A OR B) and (A AND B).
4. **Node + capacitor.** The gate outputs join at one node. The capacitor smooths it and the time code Y presents a defined HIGH to the flip-flop's D pin.
5. **Clock edge.** Every 100 ms the rising edge on CLK makes the 74HC74 latch whatever is on D.
6. **Result.** Q = SELECT = 1 means “accept: x is in L”. Q = 0 means “no witness found”. Q-bar shows the opposite polarity for observation.

4. Why the error is one-sided

The caption says Q = 1 only if both conditions hold, which collapses to **A AND B** (since A AND B already implies A OR B). So the circuit accepts only when the randomized driver finds a witness:

Case	A	B	SELECT (Q)	Meaning
x not in L	always 0	any	0	Never a false accept. No error possible.
x in L, witness found	1	1	1	Correct accept (probability ≥ 1/2).
x in L, witness missed	0	1	0	False reject (probability ≤ 1/2). The only error type.

This matches the footer of the diagram: *an instance outside L never gives SELECT = 1*. A wrong answer can only be a missed YES, never a fake YES.

5. Repeating trials to push the error down

One trial can wrongly reject a YES instance with probability up to 1/2. Because trials are independent and a single success is conclusive, you run the circuit k times and accept if *any* trial gives SELECT = 1. The chance of every trial failing is at most $(1/2)^k$. At the 10 Hz clock, k trials take k × 100 ms.

Trials k	Worst-case error	Time at 10 Hz
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.000976562	1.0 s
20	9.54e-07	2.0 s

6. Things to be aware of in the drawing

The figure is a conceptual illustration, not a build-ready schematic. If you ever wire it up, check these points:

- **How the two gates combine.** AND implies OR, so requiring both conditions, (A OR B) and (A AND B), reduces to A AND B. Tied together as drawn, the two outputs would instead behave like a plain OR (A OR B). The caption's “both conditions” reading is the one that keeps the logic

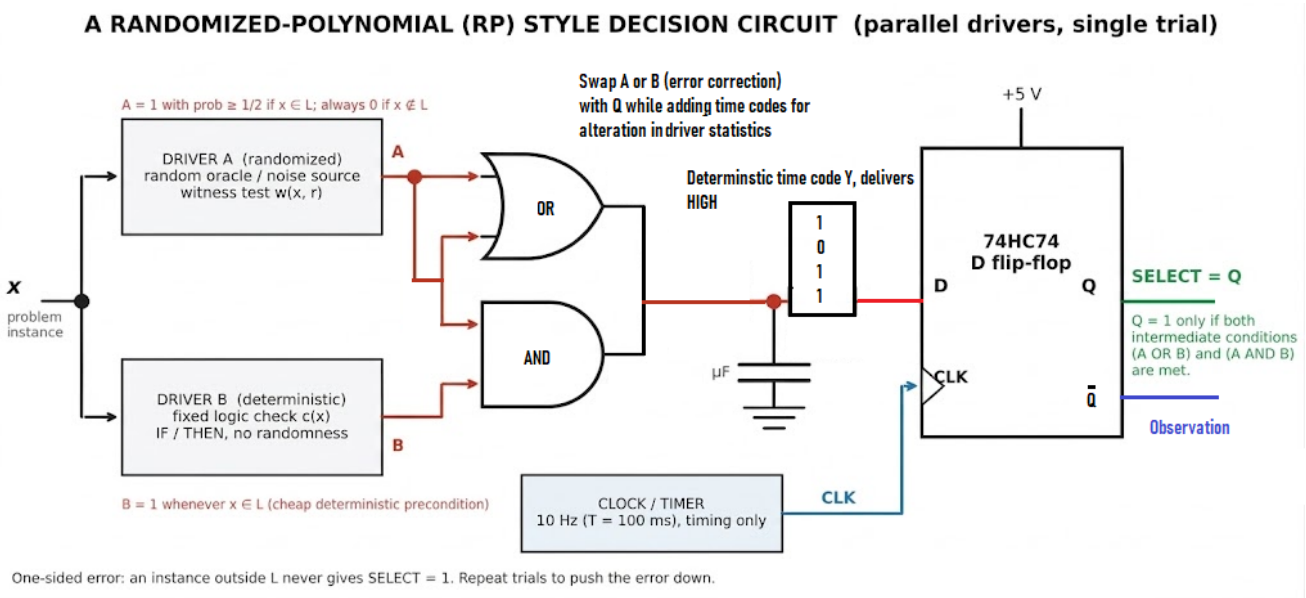
meaningful.

- **Tied gate outputs.** Directly joining two CMOS outputs causes bus contention when they disagree. A real design would use an AND gate (or a proper combining stage) to merge them.
- **A OR B alone would break the guarantee.** If B can be 1 for an x outside L, accepting on A OR B could give a false accept. Only the AND path preserves one-sided error.
- **74HC74 housekeeping.** Real use needs the PRE and CLR pins tied high, VCC and GND connected, and decoupling near the chip. These are not shown.
- **Q-bar is not extra information.** It is simply NOT Q, useful for an indicator LED or active-low logic.

Summary: two drivers test x in parallel, the gates demand agreement, a clocked D flip-flop latches the verdict, and repetition drives the false-reject rate toward zero while false accepts stay impossible.

From Electrons to Photon Statistics

Re-reading the RP-style decision circuit as a model of how electron handling shapes the light an optoelectronic device emits or detects



The source circuit: parallel drivers A and B, OR/AND gating, time code Y and a 74HC74 flip-flop

Description of the circuit. Driver A is a randomized witness test: if x is in L it outputs 1 with probability at least 1/2, and if x is not in L it always outputs 0. Driver B is a deterministic check that outputs 1 whenever x is in L. The two outputs go to an OR gate and an AND gate, meet at a node with a capacitor, and pass through the fixed time code Y = 1011 to the D input of a 74HC74 flip-flop. A 10 Hz clock tells the flip-flop when to sample, and Q (SELECT) holds the one-bit verdict, with Q-bar as the observation output. Because A is always 0 outside L, SELECT can never be 1 for a NO instance: the error is one-sided, and repeating trials shrinks the remaining error as $(1/2)^k$.

1. The claim

In electronics and optoelectronics, **the way electrons are manipulated determines the statistical distribution of the photons emitted or processed**. If electrons arrive at a junction as a random, uncorrelated stream, the photons come out random too. If the electrons are metered out one at a time, the photons inherit that regularity. The RP circuit is a convenient frame for this: it has a *random* branch, a *deterministic* branch, a gating stage, a smoothing capacitor, a timing code and a clocked latch. Each has a physical counterpart in a light-emitting or light-detecting device.

2. Three numbers that describe photon statistics

Quantity	Definition	Reading
Fano factor F	F = Var(n) / ⟨n⟩ for photons counted in a window	F = 1 Poissonian (laser, random stream). F > 1 super-Poissonian (thermal, bunched). F < 1 sub-Poissonian (regulated, antibunched).
Mandel Q	Q = F − 1	Q = 0, > 0, < 0 for the three cases above. Only Q < 0 has no classical-wave explanation.
$g^{(2)}(0)$	Probability of two photons arriving together, relative to chance	Thermal light: 2. Laser: 1. Ideal single-photon emitter: 0.

3. Re-labelling the circuit

Circuit element	Electronic / optoelectronic counterpart	Statistical effect
Input x	The excitation condition: bias, injected carriers, optical pump	Sets the mean photon number.
Driver A (random oracle / noise source)	Uncorrelated carrier injection: shot noise, spontaneous recombination, thermal carriers	Pushes the photon count toward Poissonian or, with thermal mixing, super-Poissonian.
Driver B (deterministic check)	Regulated injection: constant-current drive through a high impedance, a single-electron turnstile, Coulomb blockade in a quantum dot	Electrons arrive in a regular pattern, so photons are sub-Poissonian.
AND gate	Coincidence condition: an electron *and* a hole must both be present before radiative recombination can occur	Photon only if both carriers are available; gates the emission.
OR gate with feedback note (“swap A or B with Q”)	Either the random or the regulated channel can trigger emission; error-correcting feedback from output to pump	Feedback that suppresses pump noise is the standard route to amplitude-squeezed light.
Capacitor node (µF)	Junction or circuit capacitance with series resistance; charging energy in a single-electron device	Sets the bandwidth over which noise is smoothed or blocked.
Time code Y = 1011	A gated pump-pulse pattern: excite in time bins 1, 3 and 4, skip bin 2	Fixes *when* photons may appear, shaping the temporal statistics.
Clock, 10 Hz	Pulsed excitation or detector gating (real devices: kHz to GHz)	Defines the counting window; 10 Hz is symbolic.
74HC74 D flip-flop, Q = SELECT	Photodetector plus threshold discriminator and latch: a “click”	Converts the photon stream into a recorded 0/1 outcome per window.
Q-bar (Observation)	The complementary “no click” record	Same information, opposite polarity.

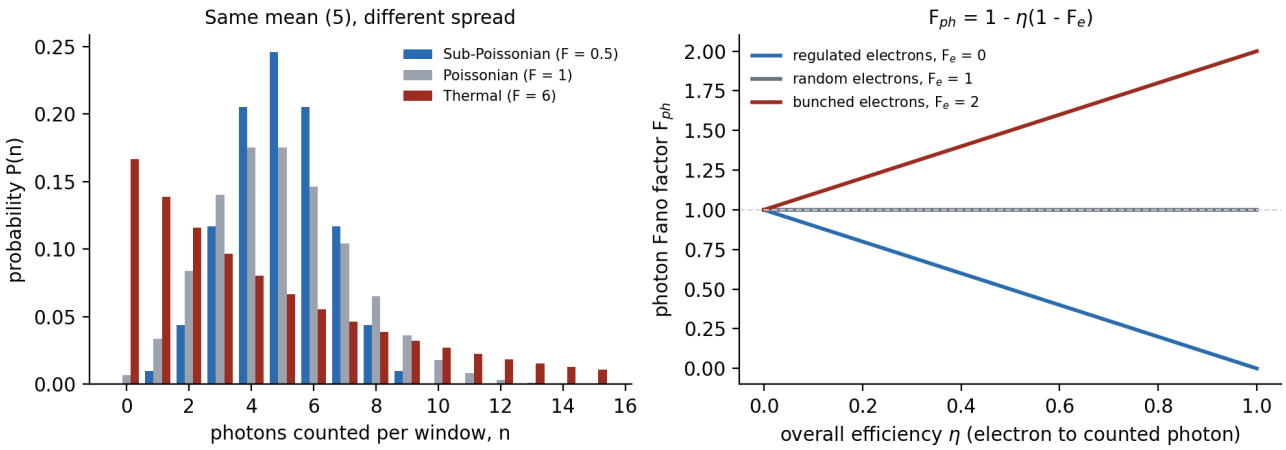
4. How electron handling sets the photon distribution

Suppose each injected electron produces a counted photon with overall probability η (internal quantum efficiency times extraction times detector efficiency), independently of the others. Random thinning of a stream gives a simple relation between the electron Fano factor F_e and the photon Fano factor F_{ph}:

$$\mathbf{F_{ph} = 1 - \eta\,(1 - F_e)}$$

Three regimes follow directly:

- **Random electrons ($F_e = 1$)**, the Driver A case: $F_{ph} = 1$ whatever η is. The photons are Poissonian.
- **Regulated electrons ($F_e = 0$)**, the Driver B case: $F_{ph} = 1 - \eta$. With $\eta = 0.8$ the photon stream has $F = 0.2$, strongly sub-Poissonian. The regularity survives only as far as the efficiency allows.
- **Bunched electrons ($F_e > 1$)**, for example trap-limited or avalanche-like injection: $F_{ph} > 1$, super-Poissonian.



Left: photon-number distributions with the same mean. Right: the Fano-factor relation above.

5. One-sided error becomes detector behaviour

The circuit's guarantee was: *no witness, no accept*. A photon detector behaves the same way. With no photon present there is (apart from dark counts) no click, so false accepts are essentially absent. A photon that is present can still be missed, with probability $1 - \eta$, which plays the role of the false-reject chance ($\leq$ 1/2 in the RP definition, so it needs $\eta \geq 1/2$ to match).

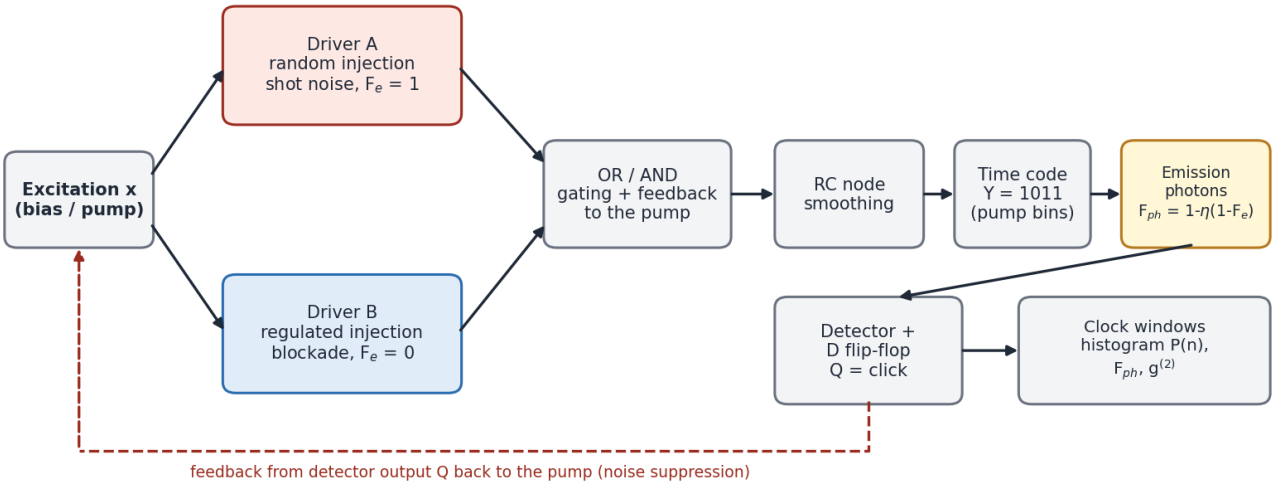
Repeating trials has a direct counterpart too: counting over k clock windows drives the chance of missing a real signal down to $(1 - \eta)^k$. With $\eta = 0.5$, ten windows leave about 0.1%. Dark counts are the one place the analogy breaks cleanly: they are false accepts, which the ideal RP circuit never has, and they accumulate as you add windows.

6. Walking through the re-labelled circuit

1. **Excite.** The bias or pump (x) goes to both the random and regulated electron channels.
2. **Inject.** Channel A delivers carriers at random times. Channel B delivers them in a metered, nearly regular pattern.
3. **Gate.** The AND stage lets recombination proceed only when carrier pairs coincide. The feedback path trims pump fluctuations.
4. **Smooth and time.** The RC node filters fast noise and the pulse pattern 1011 fixes the allowed emission bins.
5. **Detect and latch.** On each clock edge the detector stage records click or no click.
6. **Accumulate.** The histogram of clicks over many windows is the photon-number distribution, and its Fano factor reveals which electron statistics were in charge.

7. Integration: one chain from electrons to counted photons

Putting the circuit and the physics together gives a single chain. The excitation feeds a random channel and a regulated channel in parallel. Their outputs are gated, smoothed by the RC node and sliced into time bins by the code Y. The resulting electron stream drives emission, and the photons are then thinned by the overall efficiency η before the detector and flip-flop turn them into clicks. A feedback path from the output back to the pump closes the loop, which is how pump-noise suppression is achieved in practice.



Integrated block diagram: red path = random branch and feedback, blue = regulated branch

Operating mode	Which branch dominates	Photon statistics out
A only	Random injection, no regulation	$F_{ph} = 1$ (Poissonian), or above 1 with thermal mixing.
B only	Regulated injection	$F_{ph} = 1 - \eta$ (sub-Poissonian).
A and B with AND gating	Photon only when both conditions coincide	Intermediate: noise from A is limited by the regulated gate from B.
A and B with feedback via Q	Output corrects the pump	F_{ph} pushed below 1, limited by η and loop bandwidth (set by the RC node).

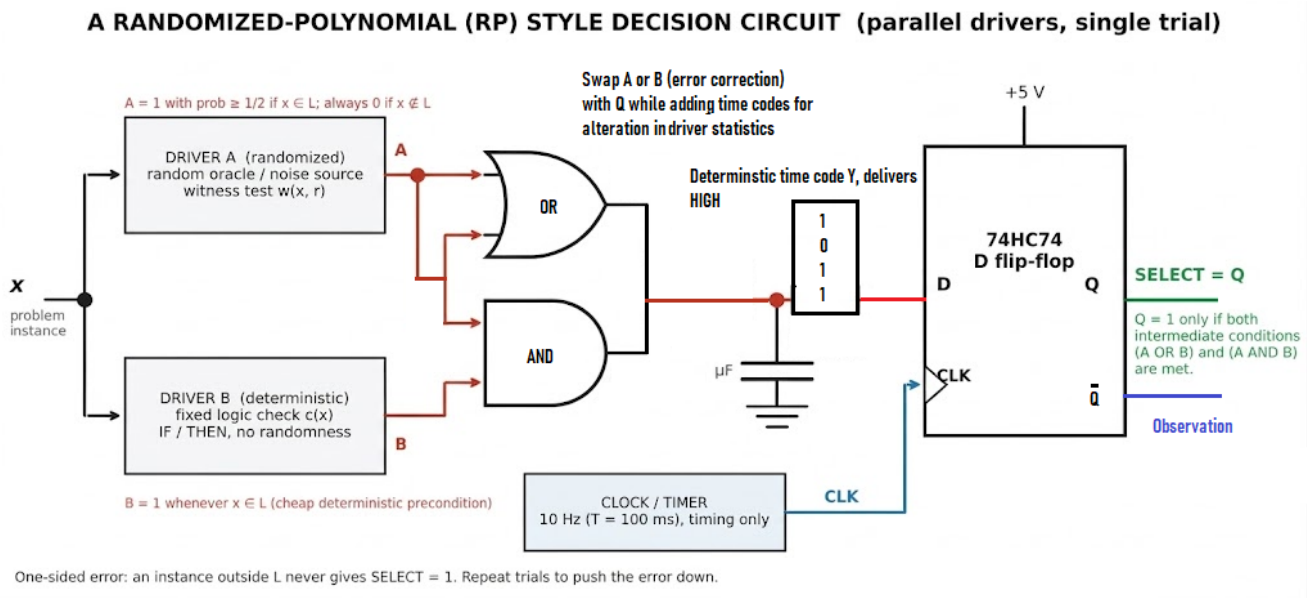
8. Limits of the analogy

- The circuit is a *decision* device with a single bit out. Photon statistics are a property of many windows, so the mapping is conceptual, not a circuit you could build to emit light.
- The relation $F_{ph} = 1 - \eta(1 - F_e)$ assumes independent loss and one photon per electron. Real devices add correlations from recombination dynamics, cavity effects and detector dead time.
- The drawing's gate wiring (OR and AND outputs tied together) is ambiguous: as a plain OR it would allow false accepts. Only the reading in which both conditions must hold, which reduces to A AND B and keeps the one-sided guarantee, is used here.
- Strongly sub-Poissonian emission in practice needs low temperature, high efficiency and low noise. Real LEDs and lasers usually sit near F = 1.

Summary: random electrons give random photons, regulated electrons give regular photons, and every loss or gate between the two pulls the distribution toward Poisson. The circuit's random driver, deterministic driver, gating, smoothing, timing and latch map onto exactly those steps.

The Protocol: Energy-Level Switching, Measurement and Closed Timelike Curves

How one trial of the RP-style circuit works as a physical protocol, and what its feedback loop has to do with time loops



The circuit this protocol is built on

1. The protocol in five steps

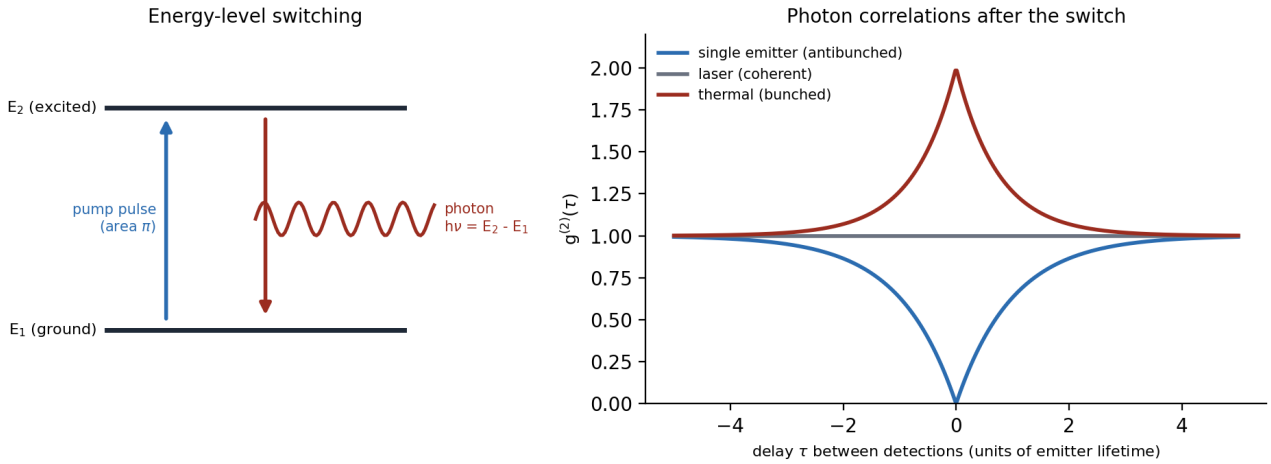
Step	Circuit element	Physical event	Result
1. Prepare	Input x, clock, time code Y = 1011	Choose the excitation and the allowed time bins (pump in bins 1, 3, 4; skip bin 2).	Defines when and how strongly electrons are driven.
2. Switch levels	Drivers A and B	An electron is driven from E_1 to E_2. On relaxation it emits a photon with $h\nu = E_2 - E_1$.	Photon emission, random (A) or metered (B).
3. Gate	OR / AND gating	Emission or counting is allowed only when the required conditions coincide.	Combined condition, read here as A AND B.
4. Measure	RC node, 74HC74, CLK	A detector turns photons into a click. On the clock edge the flip-flop latches it.	Q = SELECT = 1 (accept) or 0.
5. Repeat	Clock windows	Run k independent trials; accept if any trial gives Q = 1.	Worst-case error $(1/2)^k$.

2. Photons controlled by electron energy-level switching

A photon is emitted when an electron drops between two energy levels, and its energy is fixed by the gap. What the circuit's two drivers control is *when* the switch happens, and that sets the photon statistics.

- **Driver A, random excitation.** Thermal or noisy pumping makes each emitter switch at a random time. Independent emitters give Poissonian counts (Fano factor F = 1). Thermal fluctuations on top push F above 1.
- **Driver B, deterministic excitation.** A resonant pulse with area π inverts a two-level system with probability close to 1, since the excited-state probability is $\sin^2(\theta/2)$. After emitting, the emitter is in its ground state and cannot emit again until it is re-excited, so photons tend to arrive one at a time. This is antibunching, with $g^{(2)}(0)$ approaching 0 and F < 1.

- **Clock and time code.** They decide which bins are pumped, so they fix the counting window and the temporal pattern of the emission.



Left: the pump moves an electron from E_1 to E_2, and relaxation emits a photon. Right: the correlation function $g^{(2)}(\tau)$ for three kinds of source.

3. What measurement does

Photon-number outcomes follow the Born rule, $P(n) = |\langle n|\psi\rangle|^2$. Right after emission the emitter and the photon can be entangled, so detecting the photon also updates what you know about the emitter. The measurement chain adds its own effects:

- A detector with efficiency η thins the stream, so $F_{ph} = 1 - \eta(1 - F_e)$. Regularity from Driver B survives only as far as η allows.
- The click is an irreversible, thresholded event. That is why a clocked flip-flop is a fair model: it samples at the clock edge and holds the result.
- Dark counts (clicks with no photon) are false accepts. The ideal RP circuit has none, so dark counts are where the analogy breaks.

4. The feedback loop and closed timelike curves

The diagram's note, “swap A or B with Q while adding time codes”, describes feeding the output Q back into the drivers. That loop is what connects the circuit to the idea of a **closed timelike curve (CTC)**: a worldline that returns to its own past. CTCs appear as solutions of general relativity (for example Gödel's rotating universe and some wormhole geometries). There is no evidence that they exist, and Hawking's chronology protection conjecture argues that physics forbids them.

Deutsch's model. In David Deutsch's 1991 quantum model, a system going around a CTC must come back in the same state it left in. The state ρ of the looping system satisfies a fixed-point condition:

$$\rho_{CTC} = \mathrm{Tr}_{other}\left[U\left(\rho_{other} \otimes \rho_{CTC} \right) U^{\dagger} \right]$$

Grandfather paradox as a bit. If the loop must flip a bit (x = NOT x), no definite value is consistent. Deutsch's rule picks the 50/50 mixture, which is the only self-consistent state. It echoes the 1/2 success threshold of the randomized driver, but this is an analogy and not a derivation.

Complexity. Aaronson and Watrous (2009) showed that with CTCs, both classical and quantum computers would solve exactly the problems in PSPACE, a class that contains RP. **Experiments.** Ringbauer and colleagues (2014) simulated CTC behaviour with photons using post-selection. This reproduces the mathematics, not an actual loop through time.

In the circuit, the feedback from Q to the pump is an ordinary electrical loop, not a time loop. It resembles the CTC fixed-point condition only in that the output must be consistent with the input it helps produce.

5. Parameters of the system

Parameter	Role	Where it appears
$E_2 - E_1$	Photon energy and wavelength	Level switching

Parameter	Role	Where it appears
τ (emitter lifetime)	Emission rate; recovery time between photons	$g^{(2)}(\tau)$ dip width
Pulse area θ	How deterministic the excitation is (π gives inversion)	Driver B
F_e, η	Electron noise and overall efficiency, giving F_{ph}	Sections 2 and 3
Dark-count rate	Sets the false-accept rate	Detector and flip-flop
Clock period T	Counting window; 100 ms here, ns or shorter in real devices	CLK
Time code Y	Which bins are pumped	Pump timing
k (trials)	Number of repetitions; error $(1/2)^k$	Repeat step
Loop consistency	Fixed-point condition on the fed-back state	CTC reading of the feedback

6. What is established and what is speculative

Status	Content
Established physics	Level-switching emission; Poissonian, thermal and antibunched statistics; the Born rule; the thinning relation for detector efficiency; one-sided error and $(1/2)^k$ repetition.
Model-dependent	Deutsch's fixed-point rule for CTCs. Other models (post-selected CTCs, for example) give different predictions, and Deutsch's model has been criticised for its nonlinear features.
Speculative	Whether CTCs exist at all, and any link between the circuit's feedback and a real time loop.

7. References to check

- D. Deutsch, “Quantum mechanics near closed timelike lines”, Physical Review D, 1991.
- S. Aaronson and J. Watrous, “Closed timelike curves make quantum and classical computing equivalent”, Proceedings of the Royal Society A, 2009.
- M. Ringbauer et al., “Experimental simulation of closed timelike curves”, Nature Communications, 2014.
- S. Hawking, “Chronology protection conjecture”, Physical Review D, 1992.

These citations are from memory and have not been checked against the publications.